Experiment 01: Determination of the molecular weight of a substance by cryoscopic method.

□ Define colligative properties. Mention some of them.

- The properties that depend only upon the number of particles (ions or molecules) that are dissolved in the solvent and not on the identity of the particles, are colligative properties. Colligative properties include:

(i) vapor pressure lowering
(ii) Boiling point elevation
(iii) Freezing point depression
(iv) Osmotic pressure.

□ Why colligative properties of electrolytes and non-electrolytes are not same?

- Colligative properties depend on the solute particles, not on their nature. Electrolytes dissociate into ions, increasing particle count and significantly enhancing colligative effect having a large number of dissolved particles. But non-electrolytes remain as molecules showing poor colligative effe

□ Explain the term "Depression of freezing point."

- When a solute is added to a solvent, the freezing point of the solution decreases compared to the pure solvent. This is because the solute particles interfere with the solvent molecules' ability to form lattice structure. The more solute particles present, the greater the depression of freezing point.

□ Write equation showing relation between molecular mass of a non-volatile and non-electrolyte solute and depression of freezing point.

We know,
Depression of freezing point,

$$\Delta T = T_f^\circ - T_f = K_f m$$

$$\Rightarrow \Delta T = K_f \cdot \frac{1000\,\omega_1}{M\omega_2} \qquad \left[m = \text{molality} = \frac{1000\,\omega_1}{M\omega_2}\right]$$

$$\therefore M = \frac{K_f \times \omega_1 \times 1000}{\Delta T\, \omega_2}$$

where,

ΔT = Depression of freezing point
K_f = Freezing point depression constant / Cryoscopic constant
ω_1 = wt. of solute
ω_2 = wt. of solvent
M = Molecular mass of solute

□ When ΔT_f will be equal to K_f?

We know,

$\Delta T_f = K_f m$ where,

ΔT_f = Depression of freezing point
K_f = Freezing point constant / Cryoscopic constant
m = molality

When, $m = 1$ i.e. for one molal solution, $\Delta T_f = K_f$. It denotes that for one molal solution, Freezing point depression and Cryoscopic constants are equal.

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□ What is the freezing point of the 0.33m solution of a non-volatile, non-electrolyte solute in benzene. (Fp of benzene = 5.5°C, Kf for benzene = 5.12°Cm^{-1}).

We know,

$\Delta T_f = K_f m$

$\therefore T_f^\circ - T_f = 5.12 \times 0.33$

$\Rightarrow 5.5 - T_f = 1.7$

$\Rightarrow T_f = 5.5 - 1.7$

$= 3.8°C$

Given,

$\Delta T_f = T_f^\circ - T_f$

$m = 0.33$

$T_f^\circ = 5.5°C$

$K_f = 5.12°C\,m^{-1}$

□ What is the freezing point of a solution containing 3.42g of sucrose and 5.00g of water. (MW of sucrose = 342g, K_f=1.86)

$\Delta T = K_f \cdot \frac{1000\,\omega_1}{M\omega_2}$

$\therefore \Delta T = \frac{1.86 \times 1000 \times 3.42}{342 \times 5.00}$

$= 3.72$

$\therefore T_f^\circ - T_f = 3.72$

$\Rightarrow T_f = T_f - 3.72$

$= 0 - 3.72$

$= -3.72°C$

$K_f = 1.86$
$\omega_1 = 3.42g$
$\omega_2 = 5.00g$
$M = 342g$
$T_f^\circ = 0°C$

• Electrolyte এর জন্য vant hoff vactor 'i' count করতে হয়, which is complex

• Non-volatile solute ব্যবহার করা হয় যাতে এটির volatility এর জন্য extra কোনো vapour pressure add না হয় এবং purely solvent এর vapor pressure lowering consider হয়,

• যেহেতু non-electrolyte ion এ বিভক্ত না হয়ে molecule এ dissociate হয় এবং no of molecules number of solute added, তাই calculation easy

□ Molality is the measure of the amount of solute in a solution relative to a given mass of solvent.

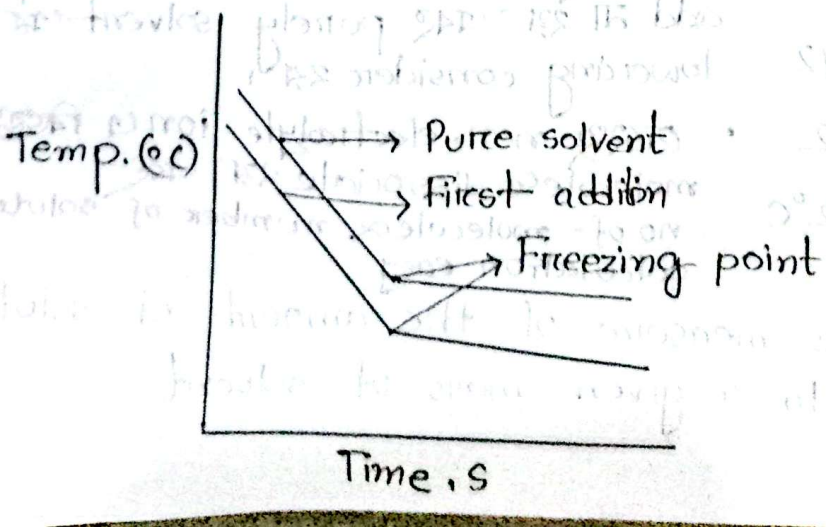
□ Why freezing point depression occurs when a non-volatile and non-electrolyte solute is added to a solution.

- Freezing point depression occurs when a non-volatile, non-electrolyte solute is added to a solvent. because the solute disrupts the solvent's ability to form a regular crystal lattice structure, lowering it's vapor pressure As, $\Delta P \propto \Delta T_f$ i.e. vapour pressure lowering is proportiond to freezing point lowering, the temperature of freezing also decreases.

□ Will the depression of freezing point of one mole of sucrose and one mole of NaCl be equal? Give reason.

- The freezing point depression of sucrose and NaCl will never be equal. NaCl is an electrolyte that dissociates into two ions (Na^+ and Cl^-) in solution, while sucrose is not electrolyte that does not dissociate. Since NaCl produces twice as many particles as sucrose in solution, it will cause a greater freezing point depression.

□ Sketch the graph to show the depression of freezing point of a solvent due to addition of a solute.

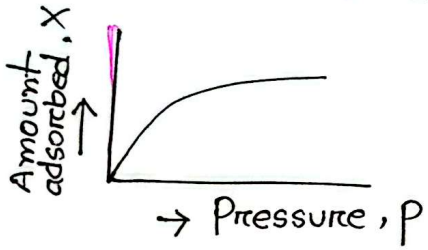


Experiment 02 : Studying the adsorption of acetic acid on charcoal and examining the validity of Freundlich's adsorption isotherm and Langmuirs adsorption isotherm.

- Adsorption depends on temperature and pressure

□ Define adsorption isotherm.

-When the temperature is mentioned constant and pressure is changed then the curve between amount of gas adsorbed and pressure is known as adsorption isotherm.



□ What is activated charcoal ?

- Activated charcoal is a form of carbon processed to have small, low volume pores that increase the surface area available for adsorption or chemical reaction.

□ Draw plots showing validity of Freundlich's adsorption isotherm and Langmuir's adsorption isotherm. Give the values of slope and intercepts of them from straight line equation.

- From Freundlich adsorption isotherm,

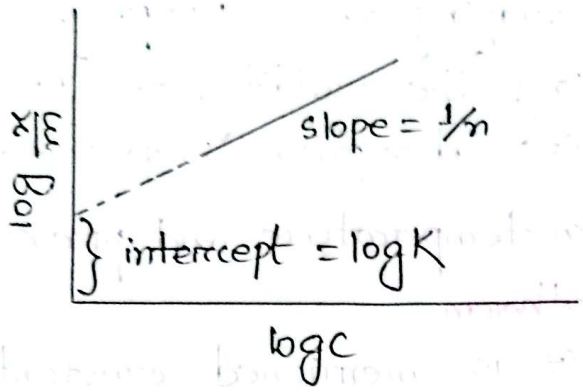
$$\frac{x}{m} = Kc^{1/n}$$

x = Amount of solute adsorbed
m = Amount of adsorbing material
c = eq. concentration of adsorbate
K = constant
n = constant, $[1/n < 1]$

Taking logarithm,

$$\log\frac{x}{m} = \log K + \frac{1}{n}\log c$$

Thus, a plot of $\log\frac{x}{m}$ vs $\log c$ should be straight line where slope $1/n$ and intercept $\log K$. [Not applicable for high pressure].



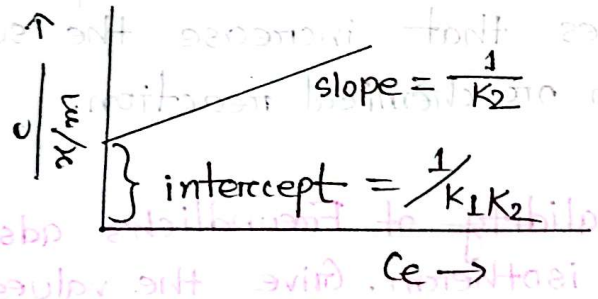
From Langmuir adsorption isotherm,

$$\frac{x}{m} = \frac{K_1 K_2 C}{1 + K_1 C}$$

By re-arranging,

$$\frac{c}{x/m} = \frac{1}{K_1 K_2} + \frac{c}{K_2}$$

Thus a plot of $\frac{c}{x/m}$ Vs ce should be straight line where slope $1/K_2$, and intercept $1/K_1K_2$.



• **Absorption:** The process by which one substance assimilated is uniformly distributed throughout the body of liquid or solid to form a solution.

• **Adsorption:** The phenomenone of concentration of molecules of gas or liquid at a solid surface is called adsorption.

• **Adsorbate:** The substance that deposits at the surface.

• **Adsorbant:** The solid on whose surface the deposition occurs.

Experiment 03: Studying the liquid-liquid miscibility of three liquid system and determining the position of one tie line.

☐ Calculate degree of freedom of three component system for both homogeneous & heterogeneous system.

According to the phase rule for three component system,

$F = C - P + 2$

$= 3 - P + 2 \quad = 5 - P$

C = component = 3
F = Degree of freedom
P = phase

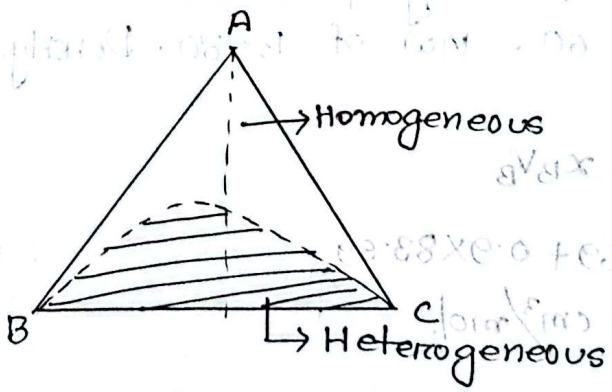
For homogeneous system, P = 1,

$F_{homo} = 5 - 1 = 4$

For heterogeneous system, P = 2,

$F_{hetero} = 5 - 2 = 3$

☐ Draw the phase diagram of three component system and show the homogenieous and heterogeneous area in the diagram.



- **Degree of freedom:** It refers to the number of independent variables that can be estimated without disturbing the no of phases in equillibrium.
- **Phase:** It is a distinct, homogeneous state of matter that is chemically and physically uniform.
- **Component** is a distinct chemical species that makes up a mixture or system.

Experiment 4 & 5 : Determination of partial molal volume of solution. (V/V) ?

Define partial molal volume.

- The partial molal volume of a substance in a mixture is a property that shows how the volume of a solution changes when a small amount of substance is added, while keeping other factors constant.

The partial molal volume V_J of a substance at same general composition is defined formally as follows;

$$V_J = \left(\frac{\partial V}{\partial n_J}\right)_{P,T,n'}$$

where, the subscript n' signifies that the amount of all other substance present are constant.

Calculate the value of Vm for the mixture of A and B having the following particulars mole fraction of A = 0.1, MW of A = 60, MW of B = 80. Density of mixture = 0.957 g/cm³

$$V_m = x_A V_A + x_B V_B$$

$$= 0.1 \times 62.69 + 0.9 \times 83.59$$

$$= 81.469 \text{ cm}^3/\text{mol}.$$

$x_A = 0.1$

$x_B = 1 - 0.1 = 0.9$

$V_A = \frac{M_A}{\rho} = \frac{60}{0.957}$

$= 62.69 \text{ gm}^3$

$V_B = \frac{M_B}{\rho} = \frac{80}{0.957}$

$= 83.59 \text{ cm}^3$

- Molarity is the number of moles of solute dissolved in 1L of solution. [Temp. dependent]
- Molality is the no. of moles of solute dissolved in 1kg of solvent. [Temp. independent]

Experiment 7:

Q Draw plots showing validity of Beer-Lambert law. Write the equation of straight line for the plot. How could you calculate unknown concentration from the plot?

- We know the Beer-Lambert law,

$$A = \varepsilon c l$$

where, A = Absorbance
c = concentration
l = thickness of medium

Ploting A against c, we get a straight line and that is,

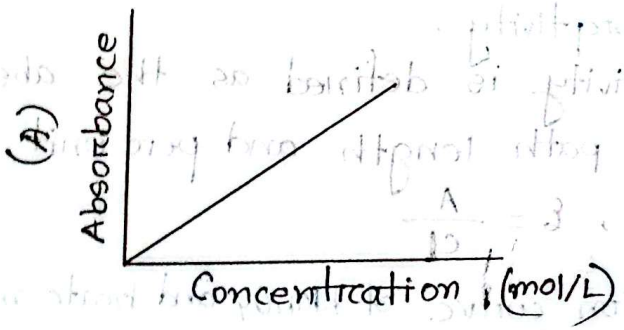
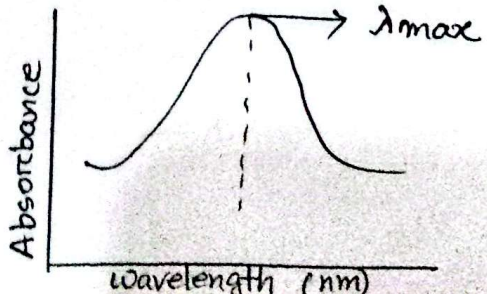


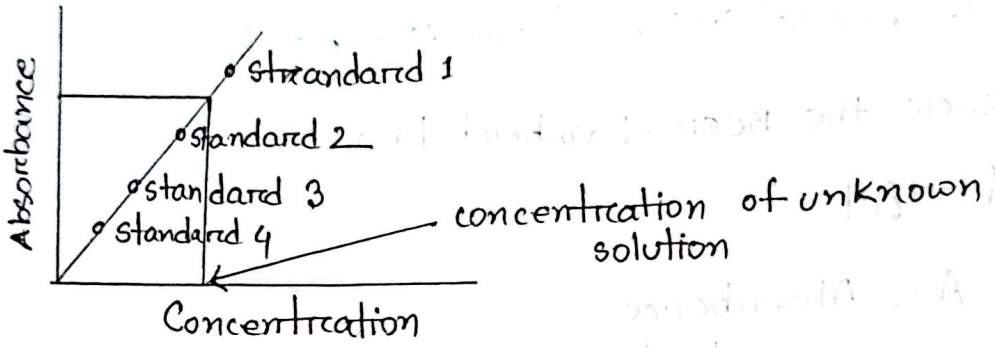
Fig: Beer-Lambert law.

There is no interaction between solute of the solution. The obtained straight line joining through the origin, therefore is the validity of Beer-Lambert law.

Calculation of unknown concentration: For measurement of concentration of unknown solution, firstly we have to find out the λmax which is obtained by absorbance vs wavelength graph.



Then the concentration of any unknown solution can be found out by measuring the absorbance at λmax and using the following plot.



From plot-

The slope, $\varepsilon = \frac{y_2 - y_1}{x_2 - x_1}$ $L\,mol^{-1}\,cm^{-1}$

সরেখ graph origin দিয়ে যায় তবে Beer–Lambert law verified.

□ Define molar absorptivity.

-Molar ~~ad~~ absorptivity is defined as the absorbance of light per unit of path length and per unit concentration.

Molar absorptivity, $\varepsilon = \frac{A}{cL}$

□ Give the absorption curve of $KMnO_4$ and locate maximum absorption

Experiment 08:

□ What is hydrolysis ? Explain by drawing graph how could you determine rate constant from a base catalyze hydrolysis of an ester.

Hydrolysis is a chemical process of decomposition in which a compound is split into other compounds by reacting with water.

Acid → 1st order

Base catalyze hydrolysis of an ester -

$$CH_3COOC_2H_5 + OH^- \rightleftharpoons CH_3COO^- + C_2H_5OH$$

Since, rate is directly proportional to both concentration of ester and alkali, the reaction is therefore second order.

$$K = \frac{1}{t} \cdot \frac{x}{a(a-x)} \quad \text{---(i)}$$

where, a = initial concentration of ester

x = concentration at time t.

Now let; C_0 and C_t are conductance at time t=0 and t=t and 'a' is the initial concentration then we get -

$$a \propto (C_0 - C_\infty) \quad \text{--- (ii)}$$

$$x \propto (C_0 - C_t) \quad \text{---- (iii)}$$

Thus, $a - x = (C_t - C_\infty)$

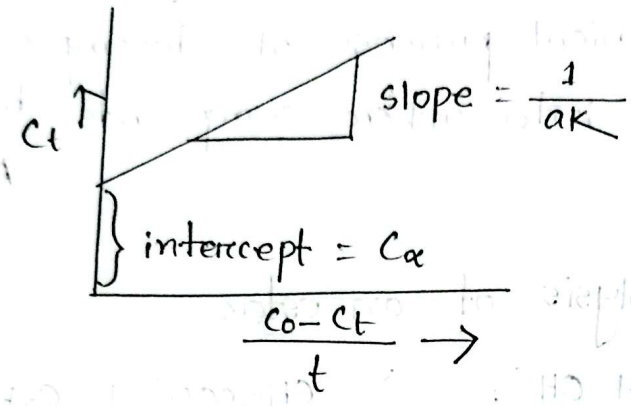
Therefore, eq (i) becomes,

$$K = \frac{1}{at} \cdot \frac{C_0 - C_t}{C_t - C_\infty}$$

By re-arranging -

$$C_t = \frac{1}{aK} \cdot \frac{C_0 - C_t}{t} + C_\infty \quad \text{---- (iv)}$$

Thus a plot of C_t vs $\frac{C_0 - C_t}{t}$ should be straight line where, slope $= \frac{1}{aK}$ and by determining slope we can determine the rate constant.



□ Write the unit of rate constant for second order reaction.

- We know, rate constant for second order reaction-

$$K = \frac{1}{t} \cdot \frac{x}{a(a-x)}$$

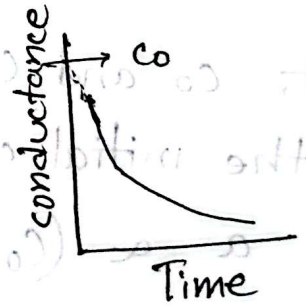
$\therefore$ Unit $= L\, mol^{-1} s^{-1}$

* First order, $K = \frac{1}{t} \ln \frac{a}{a-x}$

unit $= s^{-1}$

* Zero order, $mol\, L^{-1} s^{-1}$

* C_0 এর মান graph - 1 থেকে পাব।



* ~~Time~~ conductance 5–6 থেকে শুরু।

* weak electrolyte (CH_3COOH) hydrolysis প্রক্রিয়ায় তৈরি হওয়ায় time এর সাথে সাথে conductance বাড়ছে।

* Reaction rate constant; K is highly dependent on temp.

* Half life, $t_{1/2} = \frac{\ln K}{2}$

Experiment 09:

□ Define molar conductivity. Give unit of molar conductivity

- Molar conductivity is the molar conductance property of a solution containing one mole of the electrolyte which dissociates into ions in a volume. V.

It is obtained by multiplying specific conductance and volume,

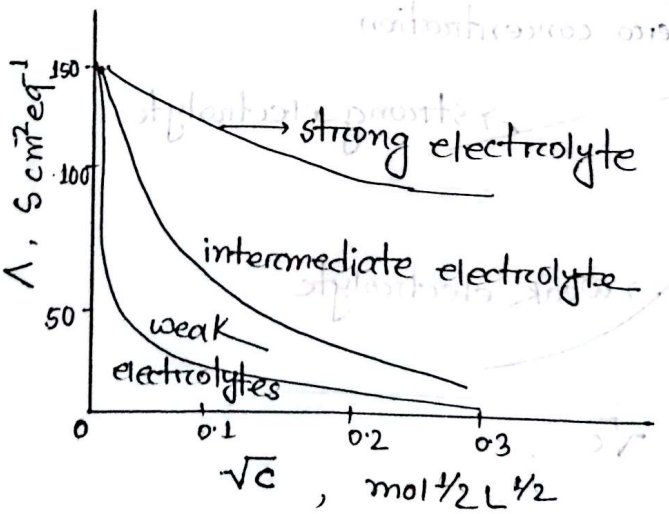
$$\Lambda_m = K \times V$$

Unit: $\Lambda_m = \frac{l}{RA} \cdot V$

$$= \frac{cm}{\Omega cm^2} \cdot \frac{cm^3}{mol} = ohm^{-1} cm^2 mol^{-1}$$

□ Why equivalent conductance increases with dilution?

- Equivalent conductance increases with dilution due to increased ionic mobility and for weak electrolytes, increased ionization. As solutions are diluted, the distance between ion increases, reducing intercionic intercactions and allowing ions to move more freely. This enhanced mobility improves electrical conductivity. Additionally, for weak electrolytes, dilution shifts the equillibrium towards the ionized form, increasing the number of ions. Thus upon dilution, equivalent conductance increases.



□ What is degree of dissociation? Write the values of k_a according to ostwalds dilution law.

- Degree of dissociation is the ratio of the conductance at any concentration and conductance at infinite dilution.

$$\alpha = \frac{\Lambda c}{\Lambda_0}$$

α = Degree of dissociation

Λc = conductance at conc. C

Λ_0 = conductance at zero concentration.

According to Ostwalds dilution law,

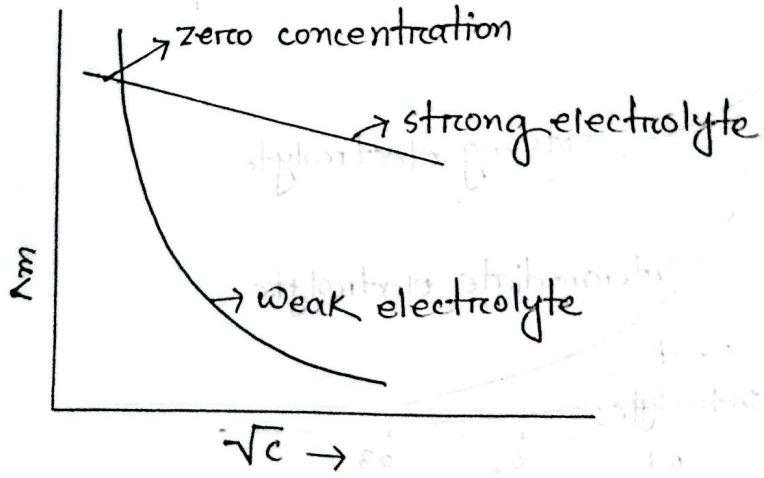
« The degree of dissociation of any weak electrolyte is inversely proportional to the sq. root of the concentration and directly proportional to the sq. root of dilution. »

According to Ostwald's law,

$$K_c = \frac{\alpha^2 c}{1-\alpha}$$

□ Draw a graph of molare conductivity Vs sq. root of concentration for weak and strong electrolyte and explain.

-



Molare conductivity of strong electrolyte:

There is only a small increase in conductance with dilution. As a strong electrolyte is completely dissociated in solution, so the number of ions remain constant on dilution. At higher concentration the greater inter ionic interactions retard the motion of ions and therefore conductance falls with increasing concentration or decreasing dilution.

Molare conductivity of weak electrolyte: [Discussed]

□ If the value for α of 0.05N acetic acid is 0.03, Calculate the value of Ka, dissociation of acid.

- From Ostwald's Law,

$$K_a = \frac{\alpha^2 C}{1-\alpha}$$

$\alpha = 0.03$
$C = 0.05$

$$= \frac{(0.03)^2 \times 0.05}{1 - 0.03}$$

$$= 1.35 \times 10^{-6}$$

□ Equivalent conductivity: Conducting powere of all ions produced by 1 gm equivalent of an electrolyte in a solution.

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